TEST REPORT

Product	:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense
Trade mark	:	Seeed Studio
Model/Type reference	:	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense
Serial Number	:	N/A
Report Number	:	EED32P80440302
Date of Issue	:	Apr. 20, 2023
Test Standards	:	ETSI EN 300 328 V2.2.2(2019-07)
Test result	:	PASS

Prepared for:

Seeed Technology Co., Ltd
9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.

Prepared by:

Centre Testing International Group Co., Ltd.
Hongwei Industrial Zone, Bao'an 70 District, Shenzhen, Guangdong, China
TEL: +86-755-3368 3668
FAX: +86-755-3368 3385

Compiled by: Frazer Li

Reviewed by: Tom Chen

Approved by: Aaron Ma

Date: Apr. 20, 2023

Check No.: 1452310323

2 Version

Version No.	Date	Description
00	Apr. 20, 2023	Original

3 Test Summary

Test Item	Test Requirement	Test Method	Limit	Result
RF output power	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.2	EN 300 328 V2.2.2 (2019-07)Clause 5.4.2	Refer clause 4.3.2.2.3	PASS
Power Spectral Density	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.3	EN 300 328 V2.2.2 (2019-07)Clause 5.4.3	Refer clause 4.3.2.3.3	PASS
Duty Cycle, Tx-sequence, Tx-gap	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.4	EN 300 328 V2.2.2 (2019-07)Clause 5.4.2	Refer clause 4.3.2.4.3	N/A[1]
Medium Utilization (MU) factor	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.5	EN 300 328 V2.2.2 (2019-07)Clause 5.4.2	Refer clause 4.3.2.5.3	N/A[2]
Adaptivity	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.6	EN 300 328 V2.2.2 (2019-07)Clause 5.4.6	Refer clause 4.3.2.6.3.2	PASS
Occupied Channel Bandwidth	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.7	EN 300 328 V2.2.2 (2019-07)Clause 5.4.7	Refer clause 4.3.2.7.3	PASS
Transmitter unwanted emissions in the out-of-band domain	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.8	EN 300 328 V2.2.2 (2019-07)Clause 5.4.8	Refer clause 4.3.2.8.3	PASS
Transmitter unwanted emissions in the spurious domain	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.9	EN 300 328 V2.2.2 (2019-07)Clause 5.4.9	Refer clause 4.3.2.9.3	PASS
Receiver spurious emissions	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.10	EN 300 328 V2.2.2 (2019-07)Clause 5.4.10	Refer clause 4.3.2.10.3	PASS
Receiver Blocking	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.11	EN 300 328 V2.2.2 (2019-07)Clause 5.4.11	Refer clause 4.3.2.11.4	PASS
Geo-location capability	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.12	EN 300 328 V2.2.2 (2019-07)Clause 4.3.2.12	Refer Clause 4.3.2.12.3	N/A[3]

Remark:

N/A[1]: Because these requirements apply to non-adaptive frequency hopping equipment mode and RF output power of greater than or equal to 10 dBm.

N/A[2] Because these requirements apply to non-adaptive frequency hopping equipment mode and RF output power of greater than or equal to 10 dBm.

N/A[3] Because these requirements apply to equipment with geo-location capability

Company Name and Address shown on Report, the sample(s) and sample Information was/ were provided by the applicant who should be responsible for the authenticity which CTI hasn’t verified.

Tx: In this whole report Tx (or tx) means Transmitter.
Rx: In this whole report Rx (or rx) means Receiver.
RF: In this whole report RF means Radiated Frequency.
CH: In this whole report CH means channel.
Volt: In this whole report Volt means Voltage.
Temp: In this whole report Temp means Temperature.
Humid: In this whole report Humid means humidity.
Press: In this whole report Press means Pressure.
N/A: In this whole report not application.

Model No.: XIAO-ESP32-S3, XIAO-ESP32-S3-Sense

XIAO-ESP32-S3-Sense is composed of XIAO-ESP32-S3 main board and expansion board. XIAO-ESP32-S3 is a single main board without an expansion board, and compared with the above-mentioned XIAO-ESP32-S3 Sense main board, there is no connector on the board. Their electrical circuit design, layout, components used and internal wiring are identical.

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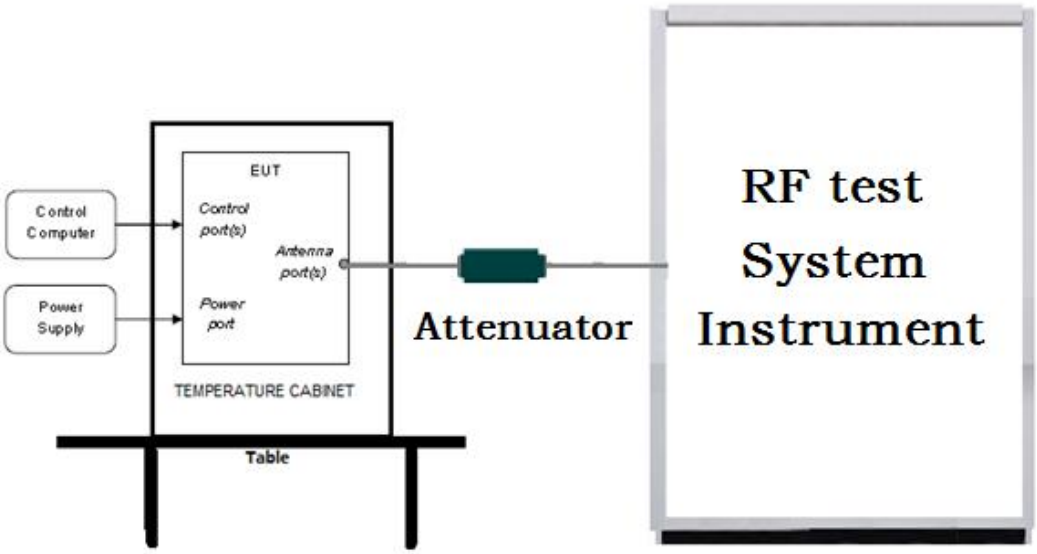
5 Test Requirement

5.1 Test setup

5.1.1 For Conducted test setup



5.1.2 For Radiated Emissions test setup

Radiated Emissions setup:

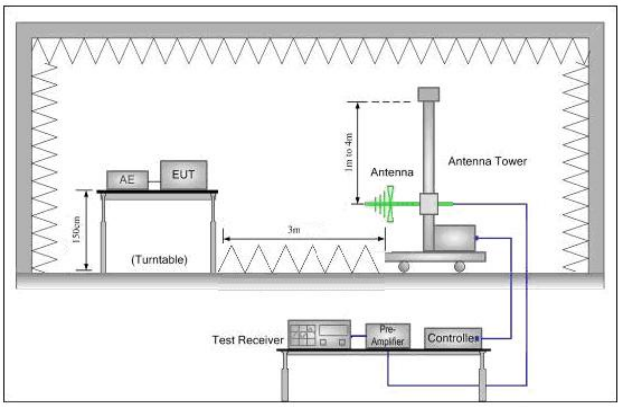


Figure 1. 30MHz to 1GHz

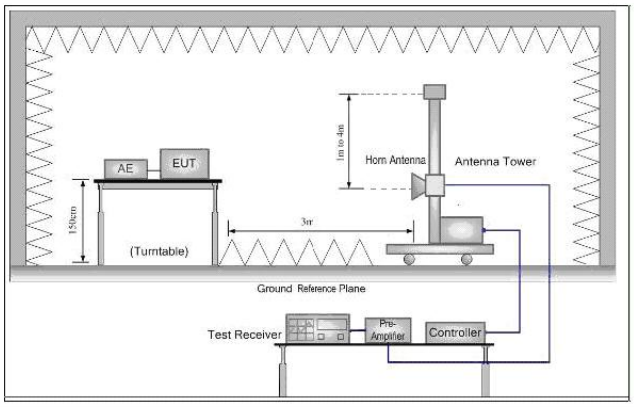


Figure 2. Above 1GHz

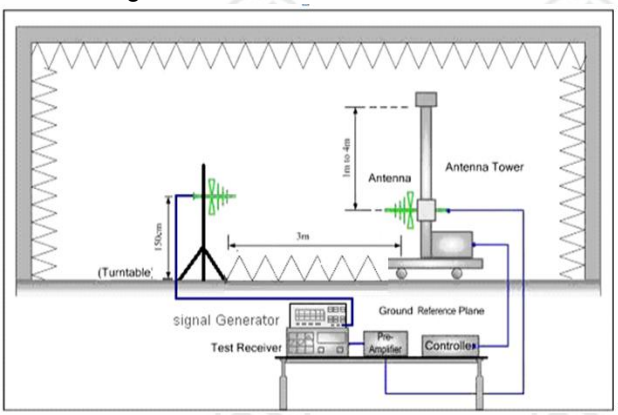


Figure 1. 30MHz to 1GHz

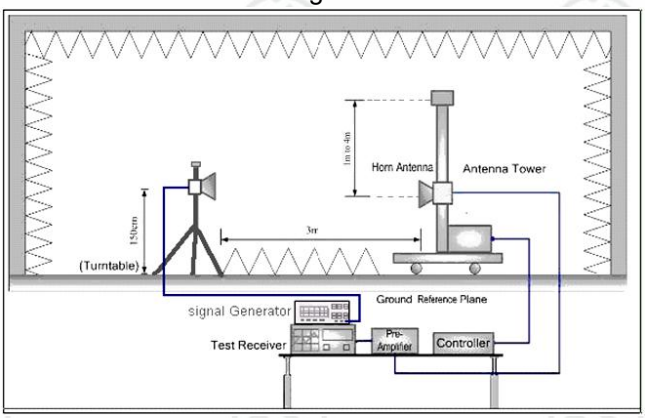


Figure 2. Above 1GHz

5.2 Test Environment

Environment Parameter	Selected Values During Tests		
Test condition	Ambient		
	Temperature(℃)	Voltage(V)	Relative Humidity%
NT/NV	24	DC 5.0	62
LT/NV	-40	DC 5.0	62
HT/NV	60	DC 5.0	62

Note:

1) The EUT just work in such extreme temperature of -40℃~+60℃, so here the EUT is tested in the temperature of -40℃~+60℃

2) NV: Normal Voltage NT:Normal Temperature
LT: Low Extreme Test Temperature HT: High Extreme Test Temperature

5.1.2 Normal test conditions

5.1.2.1 Normal temperature and humidity

Unless otherwise declared by the manufacturer, the normal temperature and humidity conditions for tests shall be any convenient combination of temperature and humidity within the following ranges:

- temperature: +15 °C to +35 °C;
- relative humidity: 20 % to 75 %.

The actual values during the tests shall be recorded.

5.1.2.2 Normal power source

The normal test voltage for the equipment shall be the nominal voltage for which the equipment was designed.

5.1.3 Extreme test conditions

Some tests in the present document need to be repeated at extreme temperatures. Where that is the case, measurements shall be made over the extremes of the operating temperature range as declared by the manufacturer.

5.3 Test Condition

Test channel

Test Mode	Tx/Rx	RF Channel		
		Low(L)	Middle(M)	High(H)
802.11b	2412MHz ~2472 MHz	Channel 1	Channel 7	Channel13
		2412MHz	2442MHz	2472MHz
802.11g	2412MHz ~2472 MHz	Channel 1	Channel 7	Channel13
		2412MHz	2442MHz	2472MHz
802.11n(HT20)	2412MHz ~2472 MHz	Channel 1	Channel 7	Channel13
		2412MHz	2442MHz	2472MHz
802.11n(HT40)	2422MHz ~2462 MHz	Channel 1	Channel 5	Channel 9
		2422MHz	2442MHz	2462MHz

Through Pre-scan all rate, 1Mbps of rate the power is the worst case of 802.11b; 6Mbps of rate the power is the worst case of 802.11g; 6.5Mbps of rate the power is the worst case of 802.11n(HT20); 13.5Mbps of rate the power is the worst case of 802.11n(HT40); only the worse case was recorded in the report.

6 General Information

6.1 Client Information

Applicant:	Seeed Technology Co., Ltd
Address of Applicant:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Manufacturer:	Seeed Technology Co., Ltd
Address of Manufacturer:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Factory:	Shenzhen Xinxian Technology Co.,Limited
Address of Factory:	F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C.

6.2 General Description of EUT

Product Name:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense	
Model No.(EUT):	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense	
Trade mark:	Seeed Studio	
Type of Modulation:	IEEE for 802.11b:DSSS(CCK,DQPSK,DBPSK) IEEE for 802.11g:OFDM(64QAM, 16QAM, QPSK, BPSK) IEEE for 802.11n(HT20 and HT40): OFDM (64QAM, 16QAM,QPSK,BPSK)	
Operating Frequency:	IEEE 802.11b/g/n(HT20): 2412MHz to 2472MHz IEEE 802.11n(HT40): 2422MHz to 2462MHz	
Channels Step:	Channels with 5MHz step	
Transmit Data Rate:	802.11b:1M/2M/5.5M/11M bps 802.11g:6M/9M/12M/18M/24M/36M/48M/54M bps 802.11n(HT20): 6.5M/13M/19.5M/26M/39M/52M/58.5M/64M bps 802.11n(HT40): 13.5M/27M/40.5M/54M/81M/108M/121.5M/135M bps	
Sample Type:	☐ Portable Use ☒ Fixed-Use ☐ Vehicle Use	
Test Power Grade:	Default	
Test Software of EUT:	ESP32S3.exe	
Antenna Type:	FPC antenna	
Antenna Gain:	2.90dBi	
Power Supply:	USB port:	DC 5.0V
Test voltage:	DC 5.0V	

6.3 Other Information

RED Directive:	2014/53/EU
Sample Received Date:	Mar. 31, 2023
Sample tested Date:	Mar. 31, 2023 to Apr. 11, 2023

Operation Frequency each of channel(802.11b/g/n HT20)

Channel	Frequency	Channel	Frequency	Channel	Frequency	Channel	Frequency
1	2412MHz	5	2432MHz	9	2452MHz	13	2472MHz
2	2417MHz	6	2437MHz	10	2457MHz		
3	2422MHz	7	2442MHz	11	2462MHz		
4	2427MHz	8	2447MHz	12	2467MHz		

Operation Frequency each of channel(802.11n HT40)

Channel	Frequency	Channel	Frequency	Channel	Frequency
1	2422MHz	4	2437MHz	7	2452MHz
2	2427MHz	5	2442MHz	8	2457MHz
3	2432MHz	6	2447MHz	9	2462MHz

6.4 Description of Support Units

The EUT has been tested with associated equipment below.

1）support equipment

Description	Manufacturer	Model No.	Certification	Supplied by
Netbook	DELL	Latitude 3490	FCC&CE	CTI

6.5 Test Location

All tests were performed at:

Centre Testing International Group Co., Ltd.

Building C, Hongwei Industrial Park Block 70, Bao'an District, Shenzhen, China

Telephone: +86 (0) 755 3368 3668 Fax:+86 (0) 755 3368 3385

No tests were sub-contracted.

6.6 Deviation from Standards

None.

6.7 Abnormalities from Standard Conditions

None.

6.8 Other Information Requested by the Customer

None.

6.9 Measurement Uncertainty (95% confidence levels, k=2)

No.	Item	Measurement Uncertainty
1	Occupied Bandwidth	0.52dB
2	RF Power conducted	0.46dB(30MHz-1GHz)
		0.55dB(1GHz-18GHz)
3	Power Spectral Density, conducted	0.57dB
4	Unwanted Emission, conducted	0.46dB(30MHz-1GHz)
		0.55dB(1GHz-18GHz)
5	All Emission, radiated	4.9dB(30MHz-1GHz)
		4.7dB(1GHz-18GHz)
6	Temperature test	0.64°C
7	Humidity test	3.8%
8	DC and low frequency voltages test	0.026%

7 Equipment List

RF test system					
Equipment	**Manufacturer**	**Model No.**	**Serial Number**	**Cal. Date (mm-dd-yyyy)**	**Cal. Due date (mm-dd-yyyy)**
Communication tset set	R&S	CMW500	107929	07-06-2022	07-05-2023
Signal Generator	R&S	SMBV100A	1407.6004K02-262149-CV	09-09-2022	09-08-2023
Spectrum Analyzer	R&S	FSV40	101200	08-01-2022	07-31-2023
RF control unit(power unit)	MWRF-test	MW100-RFCB	MW220620CTI-42	07-06-2022	07-05-2023
high-low temperature test chamber	Dong Guang Qin Zhuo	LK-80GA	QZ20150611879	12-19-2022	12-18-2023
Temperature/ Humidity Indicator	biaozhi	HM10	1804186	06-16-2022	06-15-2023
BT&WI-FI Automatic test software	MWRF-test	MTS 8310	2.0.0.0	---	---

3M full-anechoic Chamber					
Equipment	**Manufacturer**	**Model No.**	**Serial Number**	**Cal. Date (mm-dd-yyyy)**	**Cal. Due date (mm-dd-yyyy)**
RSE Automatic test software	JS Tonscend	JS36-RSE	10166	---	---
Receiver	Keysight	N9038A	MY57290136	02-27-2023	02-26-2024
Spectrum Analyzer	Keysight	N9020B	MY57111112	02-21-2023	02-20-2024
Spectrum Analyzer	Keysight	N9030B	MY57140871	02-21-2023	02-20-2024
TRILOG Broadband Antenna	Schwarzbeck	VULB 9163	9163-1148	04-28-2021	04-27-2024
Horn Antenna	Schwarzbeck	BBHA 9170	9170-832	04-15-2021	04-14-2024
Horn Antenna	ETS-LINDGREN	3117	57407	07-04-2021	07-03-2024
Preamplifier	EMCI	EMC184055SE	980597	04-20-2022	04-19-2023
Preamplifier	EMCI	EMC001330	980563	04-13-2022	04-12-2023
Preamplifier	JS Tonscend	TAP-011858	AP21B806112	07-29-2022	07-28-2023
Communication test set	R&S	CMW500	102898	12-23-2022	12-22-2023
Temperature/ Humidity Indicator	biaozhi	GM1360	EE1186631	04-11-2022 04-11-2023	04-10-2023 04-10-2024
Fully Anechoic Chamber	TDK	FAC-3	---	01-09-2021	01-08-2024
Cable line	Times	SFT205-NMSM-2.50M	394812-0001	---	---
Cable line	Times	SFT205-NMSM-2.50M	394812-0002	---	---
Cable line	Times	SFT205-NMSM-2.50M	394812-0003	---	---
Cable line	Times	SFT205-NMSM-2.50M	393495-0001	---	---
Cable line	Times	EMC104-NMNM-1000	SN160710	---	---
Cable line	Times	SFT205-NMSM-3.00M	394813-0001	---	---
Cable line	Times	SFT205-NMNM-1.50M	381964-0001	---	---
Cable line	Times	SFT205-NMSM-7.00M	394815-0001	---	---
Cable line	Times	HF160-KMKM-3.00M	393493-0001	---	---

8 Radio Technical Requirements Specification

Reference documents for testing:

No.	Identity	Document Title
1	EN 300 328 V2.2.2 (2019-07)	Wideband transmission systems; Data transmission equipment operating in the 2,4 GHz band; Harmonised Standard for access to radio spectrum

Test Results List:

EN300 328 V2.2.2		Test Descriptions & Test Conditions	Verdict	Note
Test Requirement	Test Method			
Clause 4.3.2.2	Clause 5.4.2	RF output power,		Note 1
		NT/NV	PASS	
		LT/NV	PASS	
		HT/NV	PASS	
Clause 4.3.2.3	Clause 5.4.3	Power Spectral Density		Note 1
		NT/NV	PASS	
Clause 4.3.2.4	Clause 5.4.2	Duty Cycle, Tx-sequence, Tx-gap		N/A
		NT/NV	N/A	
Clause 4.3.2.5	Clause 5.4.2	Medium Utilisation (MU) factor		N/A
		NT/NV	N/A	
Clause 4.3.2.6	Clause 5.4.6	Adaptivity (adaptive equipment using modulations other than FHSS)		Note 1
		NT/NV	PASS	
Clause 4.3.2.7	Clause 5.4.7	Occupied Channel Bandwidth		Note 1
		NT/NV	PASS	
Clause 4.3.2.8	Clause 5.4.8	Transmitter unwanted emissions in the out-of-band domain		Note 1
		NT/NV	PASS	
Clause 4.3.2.11	Clause 5.4.11	Receiver Blocking		Note 1
		NT/NV	PASS	
Clause 4.3.2.9	Clause 5.4.9	Transmitter unwanted emissions in the spurious domain		Appendix A
		NT/NV	PASS	
Clause 4.3.2.10	Clause 5.4.10	Receiver spurious emissions		Appendix A
		NT/NV	PASS	
Note 1: The test data please refer to Appendix: 2.4G WIFI of EED32P80440302				

Appendix A: Spurious emissions

Test Procedure:

1. Scan from 30MHz to 12.75GHz; find the maximum radiation frequency to measure.
2. The technique used to find the Spurious Emissions of the transmitter was the antenna substitution method. Substitution method was performed to determine the actual ERP/EIRP emission levels of the EUT.

Test procedure as below:

1) The EUT was powered ON and placed on a 1.5m hight table at a 3 meter fully Anechoic Chamber. The antenna of the transmitter was extended to its maximum length. Modulation mode and the measuring receiver shall be tuned to the frequency of the transmitter under test.
2) The EUT was set 3 meters (above 18GHz the distance is 1 meter) away from the interference-receiving antenna, which was mounted on the top of a variable-height antenna tower.
3) The disturbance of the transmitter was maximized on the test receiver display by raising and lowering from 1m to 4m the receive antenna and by rotating through 360° the turntable. After the fundamental emission was maximized, a field strength measurement was made.
4) Steps 1) to 3) were performed with the EUT and the receive antenna in both vertical and horizontal polarization.
5) The transmitter was then removed and replaced with another antenna. The center of the antenna was approximately at the same location as the center of the transmitter.
6) A signal at the disturbance was fed to the substitution antenna by means of a non-radiating cable. With both the substitution and the receive antennas horizontally polarized, the receive antenna was raised and lowered to obtain a maximum reading at the test receiver. The level of the signal generator was adjusted until the measured field strength level in step 3) is obtained for this set of conditions.
7) The output power into the substitution antenna was then measured.
8) Steps 6) and 7)were repeated with both antennas polarized.
9) Calculate power in dBm by the following formula:

   ERP(dBm) = Pg(dBm) – cable loss (dB) + antenna gain (dBd)

   EIRP(dBm) = Pg(dBm) – cable loss (dB) + antenna gain (dBi)

   EIRP=ERP+2.15dB

   where:

   Pg is the generator output power into the substitution antenna.
10) Test the EUT in the lowest channel , the Highest channel
11) Repeat above procedures until all frequencies measured was complete..

Limit:

Transmitter limits for spurious emissions

Frequency range	Maximum power, e.r.p. (≤ 1 GHz) e.i.r.p. (> 1 GHz)	Bandwidth
30 MHz to 47 MHz	-36dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87,5 MHz	-36dBm	100 kHz
87,5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36dBm	100 kHz
470 MHz to 694 MHz	-54 dBm	100 kHz
694 MHz to 1 GHz	-36dBm	100 kHz
1 GHz to 12.75 GHz	-30dBm	1MHz

Spurious emission limits for receivers

Frequency range	Maximum power e.r.p. (≤ 1 GHz) e.i.r.p. (> 1 GHz)	bandwidth
30MHz to 1GHz	-57dBm	100kHz
1GHz to 12.75GHz	-47dBm	1MHz

Radiated Spurious Emissions test Data:

1) Transmitter unwanted emissions in the spurious domain

Remark: Through Pre-scan, for 20MHz Occupied Bandwidth, 802.11 b mode was the worst case; for 40MHz Occupied Bandwidth, 802.11 n(HT40) mode was the worst case; only the worst case was recorded in the report.

With camera:

Mode:		802.11 b Transmitting						
Channel:		2412 MHz						
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.2225	150	357	-67.50	-2.09	-69.59	Pass	Horizontal
2	99.0709	150	357	-65.82	-0.99	-66.81	Pass	Horizontal
3	611.6702	150	323	-77.35	3.32	-74.03	Pass	Horizontal
4	1135.0135	150	220	-64.74	14.28	-50.46	Pass	Horizontal
5	4824.0216	150	221	-36.65	-4.72	-41.37	Pass	Horizontal
6	11023.4849	150	146	-55.98	6.01	-49.97	Pass	Horizontal
7	52.0212	150	282	-65.17	0.10	-65.07	Pass	Vertical
8	179.977	150	206	-64.35	-3.58	-67.93	Pass	Vertical
9	664.0554	150	206	-76.98	4.79	-72.19	Pass	Vertical
10	1226.2226	150	39	-65.38	15.06	-50.32	Pass	Vertical
11	4824.6716	150	86	-33.32	-4.87	-38.19	Pass	Vertical
12	10336.3891	150	312	-56.05	6.18	-49.87	Pass	Vertical

Mode:		802.11 b Transmitting						
Channel:		2472 MHz						
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	52.0212	150	349	-64.17	-2.03	-66.20	Pass	Horizontal
2	95.6756	150	357	-61.80	-1.41	-63.21	Pass	Horizontal
3	644.6535	150	291	-70.33	2.96	-67.37	Pass	Horizontal
4	1281.4281	150	31	-65.34	15.63	-49.71	Pass	Horizontal
5	4944.2796	150	107	-43.22	-4.70	-47.92	Pass	Horizontal
6	10240.8327	150	88	-55.95	5.87	-50.08	Pass	Horizontal
7	52.2152	150	321	-64.60	0.07	-64.53	Pass	Vertical
8	194.5285	150	181	-66.44	-3.89	-70.33	Pass	Vertical
9	544.9275	150	54	-72.54	1.52	-71.02	Pass	Vertical
10	1281.8282	150	62	-65.86	15.65	-50.21	Pass	Vertical
11	4944.2796	150	88	-36.83	-4.75	-41.58	Pass	Vertical
12	10299.9867	150	50	-56.48	6.30	-50.18	Pass	Vertical

Mode:	802.11 n(HT40) Transmitting							
Channel:	2422 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	52.7973	150	357	-64.73	-2.04	-66.77	Pass	Horizontal
2	100.235	150	357	-62.61	-0.87	-63.48	Pass	Horizontal
3	519.996	150	85	-73.13	0.87	-72.26	Pass	Horizontal
4	1281.2281	150	179	-65.12	15.63	-49.49	Pass	Horizontal
5	4839.6226	150	215	-41.40	-4.78	-46.18	Pass	Horizontal
6	11005.9337	150	215	-56.40	6.06	-50.34	Pass	Horizontal
7	52.0212	150	226	-63.93	0.10	-63.83	Pass	Vertical
8	95.7726	150	323	-63.86	-5.66	-69.52	Pass	Vertical
9	215.2885	150	255	-59.31	-4.77	-64.08	Pass	Vertical
10	1398.8399	150	265	-65.67	15.59	-50.08	Pass	Vertical
11	4841.5728	150	69	-36.38	-4.91	-41.29	Pass	Vertical
12	10385.7924	150	86	-55.71	6.01	-49.70	Pass	Vertical

Mode:	802.11 n(HT40) Transmitting							
Channel:	2462 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.0285	150	39	-65.67	-2.05	-67.72	Pass	Horizontal
2	96.0636	150	357	-63.98	-1.36	-65.34	Pass	Horizontal
3	613.0283	150	58	-77.75	3.32	-74.43	Pass	Horizontal
4	1281.6282	150	208	-65.86	15.63	-50.23	Pass	Horizontal
5	4909.1773	150	120	-43.51	-4.94	-48.45	Pass	Horizontal
6	9657.7439	150	349	-55.77	5.78	-49.99	Pass	Horizontal
7	52.1182	150	272	-64.68	0.08	-64.60	Pass	Vertical
8	95.4815	150	357	-63.15	-5.71	-68.86	Pass	Vertical
9	623.9904	150	272	-73.96	3.04	-70.92	Pass	Vertical
10	1253.8254	150	92	-64.76	15.36	-49.40	Pass	Vertical
11	4918.9279	150	79	-38.92	-4.92	-43.84	Pass	Vertical
12	9753.9503	150	59	-55.64	5.42	-50.22	Pass	Vertical

Without camera:

Mode:	802.11 b Transmitting							
Channel:	2412 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.3195	150	172	-67.30	-2.12	-69.42	Pass	Horizontal
2	95.3845	150	0	-64.44	-1.45	-65.89	Pass	Horizontal
3	598.9619	150	126	-69.11	2.80	-66.31	Pass	Horizontal
4	1216.6217	150	24	-64.16	15.04	-49.12	Pass	Horizontal
5	4824.0216	150	219	-29.04	-4.72	-33.76	Pass	Horizontal
6	9648.6432	150	293	-52.62	5.77	-46.85	Pass	Horizontal
7	56.2896	150	182	-65.19	-0.21	-65.40	Pass	Vertical
8	178.9099	150	349	-63.56	-3.58	-67.14	Pass	Vertical
9	646.0116	150	135	-73.58	3.26	-70.32	Pass	Vertical
10	1283.8284	150	153	-65.89	15.67	-50.22	Pass	Vertical
11	4824.0216	150	240	-38.42	-4.87	-43.29	Pass	Vertical
12	10374.0916	150	3	-56.12	6.06	-50.06	Pass	Vertical

Mode:	802.11 b Transmitting							
Channel:	2472 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	58.1328	150	246	-67.75	-2.71	-70.46	Pass	Horizontal
2	88.3028	150	11	-64.14	-3.22	-67.36	Pass	Horizontal
3	179.686	150	274	-62.87	-5.12	-67.99	Pass	Horizontal
4	1279.4279	150	226	-65.51	15.61	-49.90	Pass	Horizontal
5	4944.2796	150	178	-34.34	-4.70	-39.04	Pass	Horizontal
6	10223.9316	150	13	-55.71	5.85	-49.86	Pass	Horizontal
7	50.372	150	107	-64.85	0.31	-64.54	Pass	Vertical
8	196.2746	150	3	-64.85	-3.85	-68.70	Pass	Vertical
9	551.6212	150	152	-73.17	2.06	-71.11	Pass	Vertical
10	1278.6279	150	307	-65.45	15.63	-49.82	Pass	Vertical
11	4944.2796	150	165	-43.63	-4.75	-48.38	Pass	Vertical
12	11999.1999	150	45	-56.11	6.72	-49.39	Pass	Vertical

Mode:	802.11 n(HT40) Transmitting							
Channel:	2422 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.3195	150	78	-69.45	-2.12	-71.57	Pass	Horizontal
2	96.5487	150	346	-67.78	-1.30	-69.08	Pass	Horizontal
3	526.9807	150	3	-75.20	0.92	-74.28	Pass	Horizontal
4	1343.6344	150	355	-65.61	15.50	-50.11	Pass	Horizontal
5	4846.7731	150	207	-34.77	-4.81	-39.58	Pass	Horizontal
6	7265.5844	150	217	-50.97	-0.05	-51.02	Pass	Horizontal
7	47.6558	150	347	-64.31	-0.49	-64.80	Pass	Vertical
8	96.4516	150	198	-67.66	-5.55	-73.21	Pass	Vertical
9	670.9431	150	30	-74.38	6.10	-68.28	Pass	Vertical
10	1283.4283	150	3	-65.60	15.67	-49.93	Pass	Vertical
11	4835.7224	150	162	-44.89	-4.90	-49.79	Pass	Vertical
12	10793.3696	150	95	-56.17	6.12	-50.05	Pass	Vertical

Mode:	802.11 n(HT40) Transmitting							
Channel:	2462 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	53.7674	150	3	-71.97	-2.04	-74.01	Pass	Horizontal
2	184.4394	150	224	-68.40	-6.14	-74.54	Pass	Horizontal
3	563.7474	150	3	-75.68	1.45	-74.23	Pass	Horizontal
4	1248.2248	150	64	-65.49	15.33	-50.16	Pass	Horizontal
5	4918.9279	150	172	-36.30	-4.87	-41.17	Pass	Horizontal
6	7381.2921	150	220	-47.54	0.21	-47.33	Pass	Horizontal
7	51.0511	150	95	-66.93	0.22	-66.71	Pass	Vertical
8	180.8501	150	105	-67.20	-3.61	-70.81	Pass	Vertical
9	693.8374	150	3	-77.40	6.42	-70.98	Pass	Vertical
10	1312.8313	150	196	-66.05	15.82	-50.23	Pass	Vertical
11	4919.578	150	158	-46.00	-4.92	-50.92	Pass	Vertical
12	7380.642	150	16	-50.20	-0.09	-50.29	Pass	Vertical

2) Receiver spurious emissions test data

Remark: Through Pre-scan, for 20MHz Occupied Bandwidth, 802.11 b mode was the worst case; for 40MHz Occupied Bandwidth, 802.11 n(HT40) mode was the worst case; only the worst case was recorded in the report.

With camera:

Mode:		802.11 b Receiving						
Channel:		2412 MHz						
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.8046	150	3	-66.58	-2.21	-68.79	Pass	Horizontal
2	266.4126	150	59	-59.05	-2.82	-61.87	Pass	Horizontal
3	752.5283	150	279	-78.66	5.92	-72.74	Pass	Horizontal
4	1055.8153	150	194	-49.00	-14.48	-63.48	Pass	Horizontal
5	3186.7843	150	96	-57.99	-8.58	-66.57	Pass	Horizontal
6	9676.6338	150	279	-64.16	5.67	-58.49	Pass	Horizontal
7	36.6937	150	28	-57.48	-5.13	-62.61	Pass	Vertical
8	199.3789	150	213	-58.71	-3.80	-62.51	Pass	Vertical
9	407.6588	150	14	-65.97	0.16	-65.81	Pass	Vertical
10	1125.1438	150	161	-50.48	-14.65	-65.13	Pass	Vertical
11	3311.9281	150	88	-56.00	-9.18	-65.18	Pass	Vertical
12	9659.008	150	124	-64.45	5.17	-59.28	Pass	Vertical

Mode:		802.11 b Receiving						
Channel:		2472 MHz						
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	57.1627	150	3	-65.92	-2.50	-68.42	Pass	Horizontal
2	272.3302	150	84	-60.25	-2.31	-62.56	Pass	Horizontal
3	756.6027	150	3	-76.62	5.87	-70.75	Pass	Horizontal
4	1115.7433	150	180	-52.30	-14.57	-66.87	Pass	Horizontal
5	5024.5762	150	48	-60.01	-4.57	-64.58	Pass	Horizontal
6	9661.3581	150	267	-64.36	5.62	-58.74	Pass	Horizontal
7	54.1554	150	179	-60.11	-0.18	-60.29	Pass	Vertical
8	187.2527	150	105	-57.75	-3.87	-61.62	Pass	Vertical
9	727.9848	150	165	-78.22	9.74	-68.48	Pass	Vertical
10	1280.2515	150	191	-53.33	-12.90	-66.23	Pass	Vertical
11	3311.9281	150	90	-55.45	-9.18	-64.63	Pass	Vertical
12	9678.9839	150	68	-64.27	5.18	-59.09	Pass	Vertical

Mode:	802.11 n(HT40) Receiving							
Channel:	2422 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.3195	150	3	-65.48	-2.12	-67.60	Pass	Horizontal
2	264.4724	150	91	-60.97	-3.16	-64.13	Pass	Horizontal
3	536.5847	150	143	-70.15	1.11	-69.04	Pass	Horizontal
4	1116.3308	150	220	-48.31	-14.57	-62.88	Pass	Horizontal
5	5026.9263	150	258	-60.13	-4.56	-64.69	Pass	Horizontal
6	9665.4708	150	12	-64.22	5.64	-58.58	Pass	Horizontal
7	52.7973	150	286	-63.60	-0.01	-63.61	Pass	Vertical
8	177.2607	150	145	-58.43	-3.57	-62.00	Pass	Vertical
9	731.1861	150	286	-78.69	9.67	-69.02	Pass	Vertical
10	1179.1965	150	145	-48.86	-13.99	-62.85	Pass	Vertical
11	3311.9281	150	93	-55.41	-9.18	-64.59	Pass	Vertical
12	9659.008	150	310	-64.15	5.17	-58.98	Pass	Vertical

Mode:	802.11 n(HT40) Receiving							
Channel:	2462 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	159.993	150	3	-60.74	-6.19	-66.93	Pass	Horizontal
2	319.962	150	162	-62.66	-1.73	-64.39	Pass	Horizontal
3	695.6806	150	278	-71.08	3.16	-67.92	Pass	Horizontal
4	1191.5346	150	226	-52.79	-13.72	-66.51	Pass	Horizontal
5	3192.0721	150	3	-57.90	-8.56	-66.46	Pass	Horizontal
6	9667.8209	150	162	-64.40	5.64	-58.76	Pass	Horizontal
7	36.5967	150	190	-57.55	-5.17	-62.72	Pass	Vertical
8	197.1477	150	89	-59.88	-3.84	-63.72	Pass	Vertical
9	730.5071	150	89	-78.31	9.77	-68.54	Pass	Vertical
10	1132.7816	150	151	-51.17	-14.55	-65.72	Pass	Vertical
11	3311.9281	150	89	-55.50	-9.18	-64.68	Pass	Vertical
12	9691.9096	150	3	-64.32	5.20	-59.12	Pass	Vertical

Without camera:

Mode:	802.11 b Receiving							
Channel:	2412 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	94.7055	150	70	-67.07	-1.52	-68.59	Pass	Horizontal
2	159.993	150	3	-61.31	-6.19	-67.50	Pass	Horizontal
3	687.5318	150	158	-73.11	2.78	-70.33	Pass	Horizontal
4	1303.1652	150	269	-56.52	-12.77	-69.29	Pass	Horizontal
5	5005.7753	150	281	-60.33	-4.66	-64.99	Pass	Horizontal
6	9672.5211	150	360	-64.79	5.65	-59.14	Pass	Horizontal
7	39.992	150	0	-59.25	-3.76	-63.01	Pass	Vertical
8	199.961	150	357	-60.13	-3.79	-63.92	Pass	Vertical
9	718.4778	150	143	-76.70	9.21	-67.49	Pass	Vertical
10	1125.1438	150	239	-51.50	-14.65	-66.15	Pass	Vertical
11	5014.0007	150	216	-60.16	-4.75	-64.91	Pass	Vertical
12	9658.4204	150	265	-64.73	5.17	-59.56	Pass	Vertical

Mode:	802.11 b Receiving							
Channel:	2472 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	95.2875	150	353	-67.50	-1.46	-68.96	Pass	Horizontal
2	208.8859	150	122	-64.08	-6.34	-70.42	Pass	Horizontal
3	761.4531	150	85	-77.46	5.73	-71.73	Pass	Horizontal
4	1300.2275	150	35	-57.17	-12.74	-69.91	Pass	Horizontal
5	3192.6596	150	3	-58.93	-8.56	-67.49	Pass	Horizontal
6	9663.7082	150	158	-64.48	5.63	-58.85	Pass	Horizontal
7	49.1109	150	58	-67.81	0.03	-67.78	Pass	Vertical
8	199.961	150	357	-59.05	-3.79	-62.84	Pass	Vertical
9	730.022	150	254	-77.78	9.84	-67.94	Pass	Vertical
10	1125.1438	150	168	-52.22	-14.65	-66.87	Pass	Vertical
11	5001.0751	150	254	-60.63	-4.76	-65.39	Pass	Vertical
12	9681.3341	150	193	-64.77	5.18	-59.59	Pass	Vertical

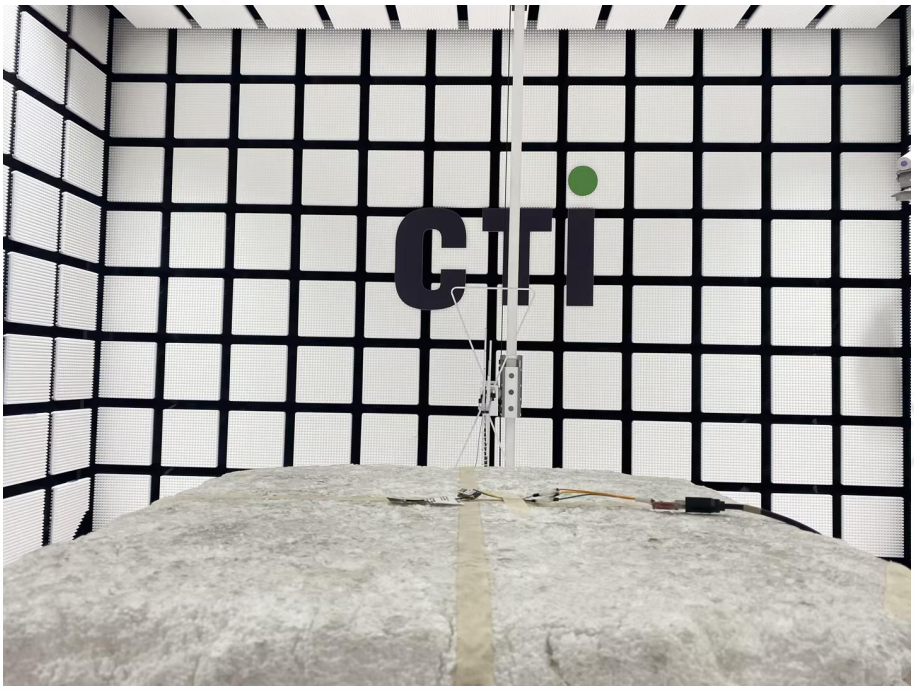
Mode:	802.11 n(HT40) Receiving							
Channel:	2422 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	57.5508	150	207	-66.72	-2.59	-69.31	Pass	Horizontal
2	265.8306	150	282	-59.26	-2.92	-62.18	Pass	Horizontal
3	536.5847	150	122	-72.88	1.11	-71.77	Pass	Horizontal
4	1176.2588	150	182	-50.08	-13.89	-63.97	Pass	Horizontal
5	3211.4606	150	122	-58.55	-8.63	-67.18	Pass	Horizontal
6	9668.996	150	207	-64.63	5.65	-58.98	Pass	Horizontal
7	40.186	150	7	-60.22	-3.67	-63.89	Pass	Vertical
8	190.454	150	55	-57.63	-3.96	-61.59	Pass	Vertical
9	727.2087	150	251	-76.00	9.70	-66.30	Pass	Vertical
10	1125.1438	150	166	-50.37	-14.65	-65.02	Pass	Vertical
11	4787.8019	150	129	-56.64	-5.24	-61.88	Pass	Vertical
12	9660.183	150	92	-64.34	5.17	-59.17	Pass	Vertical

Mode:	802.11 n(HT40) Receiving							
Channel:	2462 MHz							
NO.	Freq. [MHz]	Height [cm]	Azimuth [deg]	Level [dBm]	Limit [dBm]	Margin [dB]	Result	Polarity
1	55.7076	150	97	-65.37	-2.19	-67.56	Pass	Horizontal
2	266.2186	150	242	-58.68	-2.85	-61.53	Pass	Horizontal
3	699.2699	150	302	-72.23	3.43	-68.80	Pass	Horizontal
4	1215.6233	150	230	-49.15	-13.49	-62.64	Pass	Horizontal
5	5003.4252	150	356	-60.17	-4.68	-64.85	Pass	Horizontal
6	9656.0703	150	218	-64.32	5.61	-58.71	Pass	Horizontal
7	55.8046	150	163	-62.52	-0.24	-62.76	Pass	Vertical
8	195.9836	150	79	-59.60	-3.86	-63.46	Pass	Vertical
9	730.6041	150	127	-78.59	9.75	-68.84	Pass	Vertical
10	1280.2515	150	188	-53.32	-12.90	-66.22	Pass	Vertical
11	5009.888	150	116	-60.05	-4.75	-64.80	Pass	Vertical
12	9888.1444	150	116	-64.75	5.70	-59.05	Pass	Vertical

PHOTOGRAPHS OF TEST SETUP

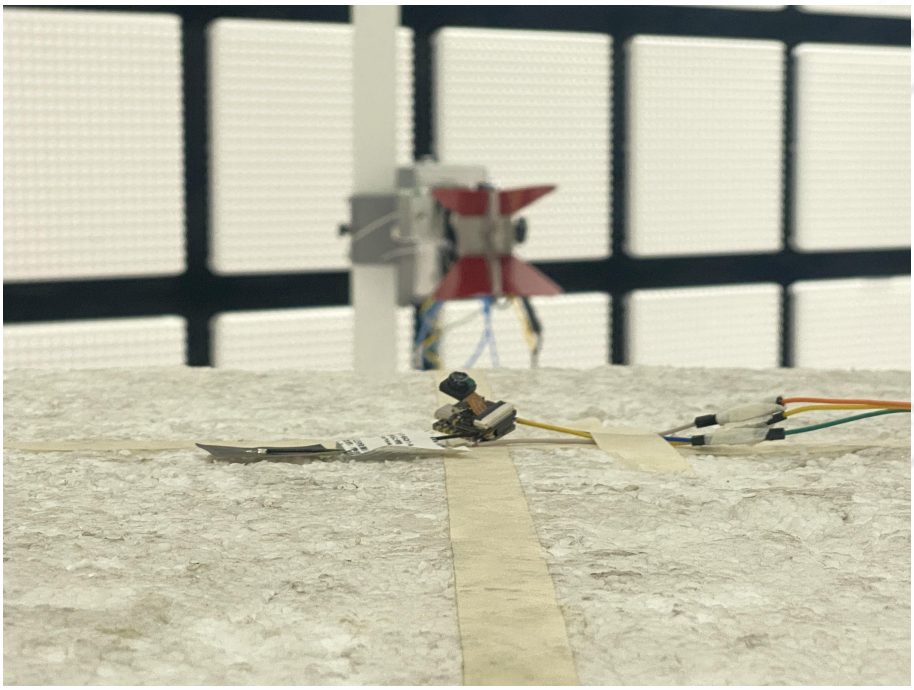
Radiated spurious emission Test Setup-1 With camera(Below 1GHz)

Radiated spurious emission Test Setup-2 With camera close up (Below 1GHz)

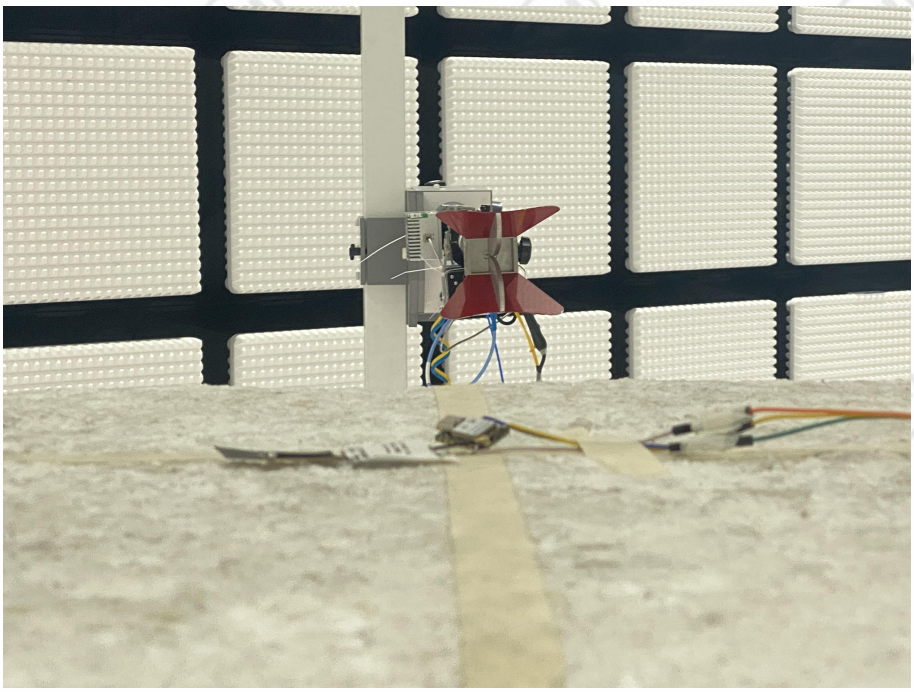


Radiated spurious emission Test Setup-3 Without camera(Below 1GHz)

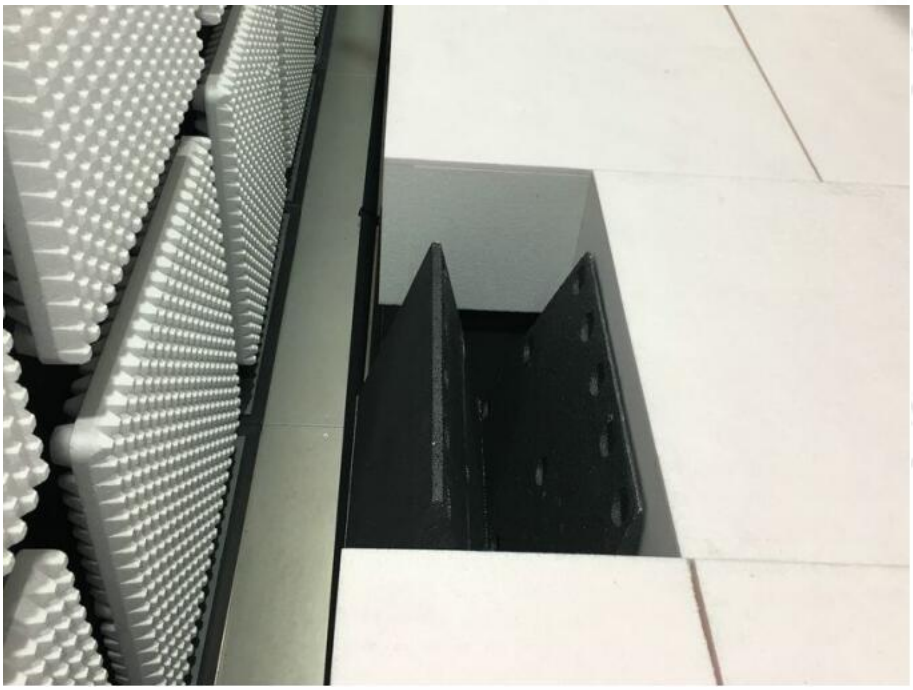
Radiated spurious emission Test Setup-4 Without camera close up (Below 1GHz)



Radiated spurious emission Test Setup-5 With camera (Above 1GHz)



Radiated spurious emission Test Setup-6 Without camera (Above 1GHz)



Radiated spurious emission Test Setup-7(Above 1GHz)

There are absorbing materials under the ground.

PHOTOGRAPHS OF EUT Constructional Details

Refer to Report No. EED32P80440301 for EUT external and internal photos.

The test report is effective only with both signature and specialized stamp, the result(s) shown in this report refer only to the sample(s) tested. Without written approval of CTI, this report can't be reproduced except in full.

*** End of Report ***